

Supporting Information

Anisotropic swelling and fracture of silicon nanowires during lithiation

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I. Experimental details

Preparation of Si nanowires. The Si nanowires were grown via a chemical vapor deposition (CVD) vapor-liquid-solid (VLS) growth method. The *n*-type Si nanowires were grown by passing 250 standard cubic centimeters (sccm) of silane (SiH_4 , 50% diluted with argon) and 100 sccm of phosphine (PH_3 , 100 ppm) at a pressure of 3 Torr across a Si substrate containing Au catalyst particles. The deposition temperature and time was 510 °C and 30 minutes. The P/Si ratio was 1.0×10^{-4} .

Construction of the all-solid Si/Li₂O/Li electrochemical device. A small piece of Si wafer containing Si nanowires emerging from the surface was cleaved off from a large Si wafer growth substrate and glued to an aluminum rod with conductive epoxy (Fig. 1a). This ensured that there was good electrical contact between the nanowires and the Si substrate, which served as the primary current collector on the Si nanowire side. Lithium metal was scratched with a tungsten wire inside a glove box filled with helium (O_2 and H_2O content < 1 ppm). Both Si and Li electrodes were mounted onto a Nanofactory® TEM-scanning tunneling microscope (STM) holder inside a glove box. Then the holder was sealed in a home-built air-tight bag filled with dry helium and transferred to the TEM. The total time of exposure to the air was less than 2 s, which limited the extent of Li_2O formation on the surface of the Li metal. The liquid cell was constructed by replacing the Li metal with a bulk LiCoO_2 counter-electrode covered by one drop of the ionic liquid electrolyte (ILE, 10% LiTFSI dissolved in P_{14}TFSI ; LiTFSI denoting lithium bis (trifluoromethylsulfonyl) imide and P_{14}TFSI denoting 1-butyl-1-methylpyrrolidinium bis (trifluoromethylsulfonyl) imide).¹

II. Supporting Figures

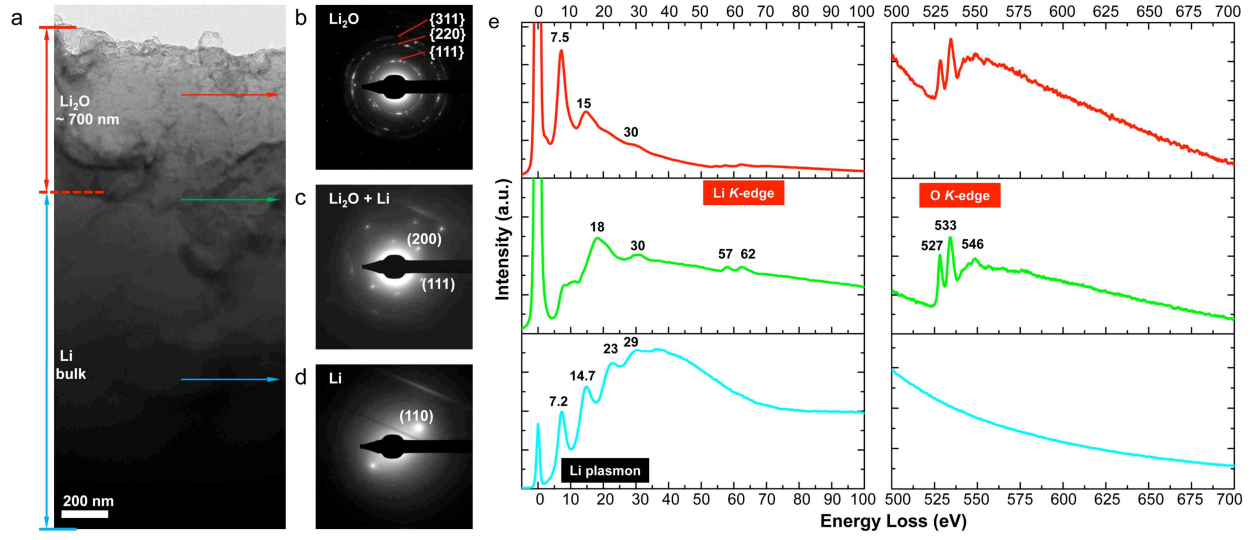


Figure S1. Characterization of the Li₂O solid electrolyte and Li electrode. (a) A typical view of the solid Li₂O/Li electrolyte/electrode. (b-d) Electron diffraction patterns (EDPs) from the Li₂O surface layer to the inner Li bulk showing the about 700-nm thick Li₂O on the Li metal. (e) Comparison of EELS spectra from the surface Li₂O layer to the inner Li region. In the low-loss region, multiple plasmon peaks were observed at 7.2, 14.4, 21.6, and 28.8 eV from the thick Li. The observed primary plasmon energy loss peak, $E_p = 7.2$ eV, agreed with the reported values for pure Li, which are in the range of 7.08 ~ 7.5 eV.² For the Li₂O thin layer, the low-loss peaks were different from those of Li metal, and the peak at 30.0 eV was relatively strong for thicker Li₂O (middle). Li K-edge peaks at 57.3 and 62.1 eV were much more obvious for Li₂O. The strong oxygen K-edge peaks with an onset around 520 eV were only visible in the Li₂O region.

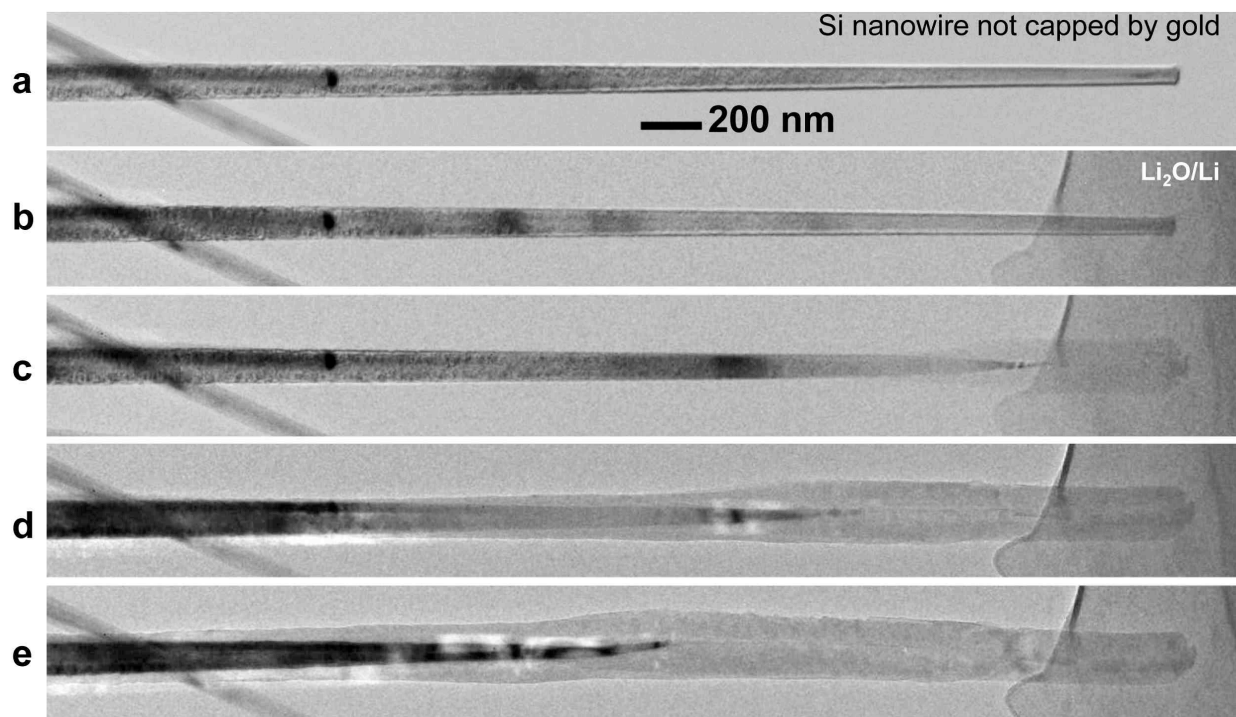


Figure S2. Similar lithiation behavior of a Si nanowire without a gold cap and directly touching $\text{Li}_2\text{O}/\text{Li}$. This indicates that the existence of a gold catalyst does not alter the lithiation behavior of the Si nanowire, which is intrinsic to the Li-Si alloying process.

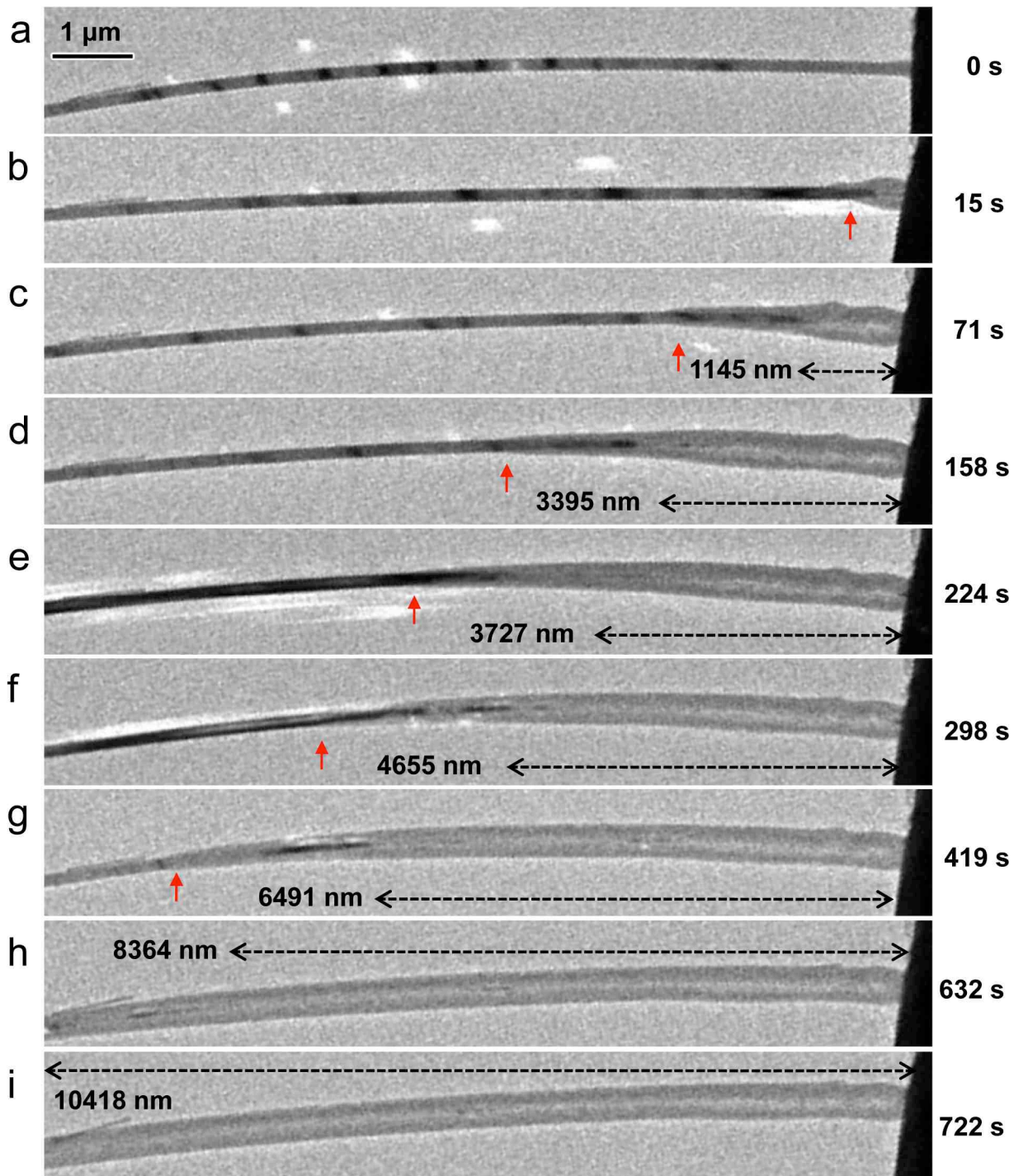


Figure S3. Crack formation during lithiation of another Si nanowire. As the reaction front (pointed out by red arrows) propagated, a crack extended following the tail of the conical residual *c*-Si core in the center of the nanowire.

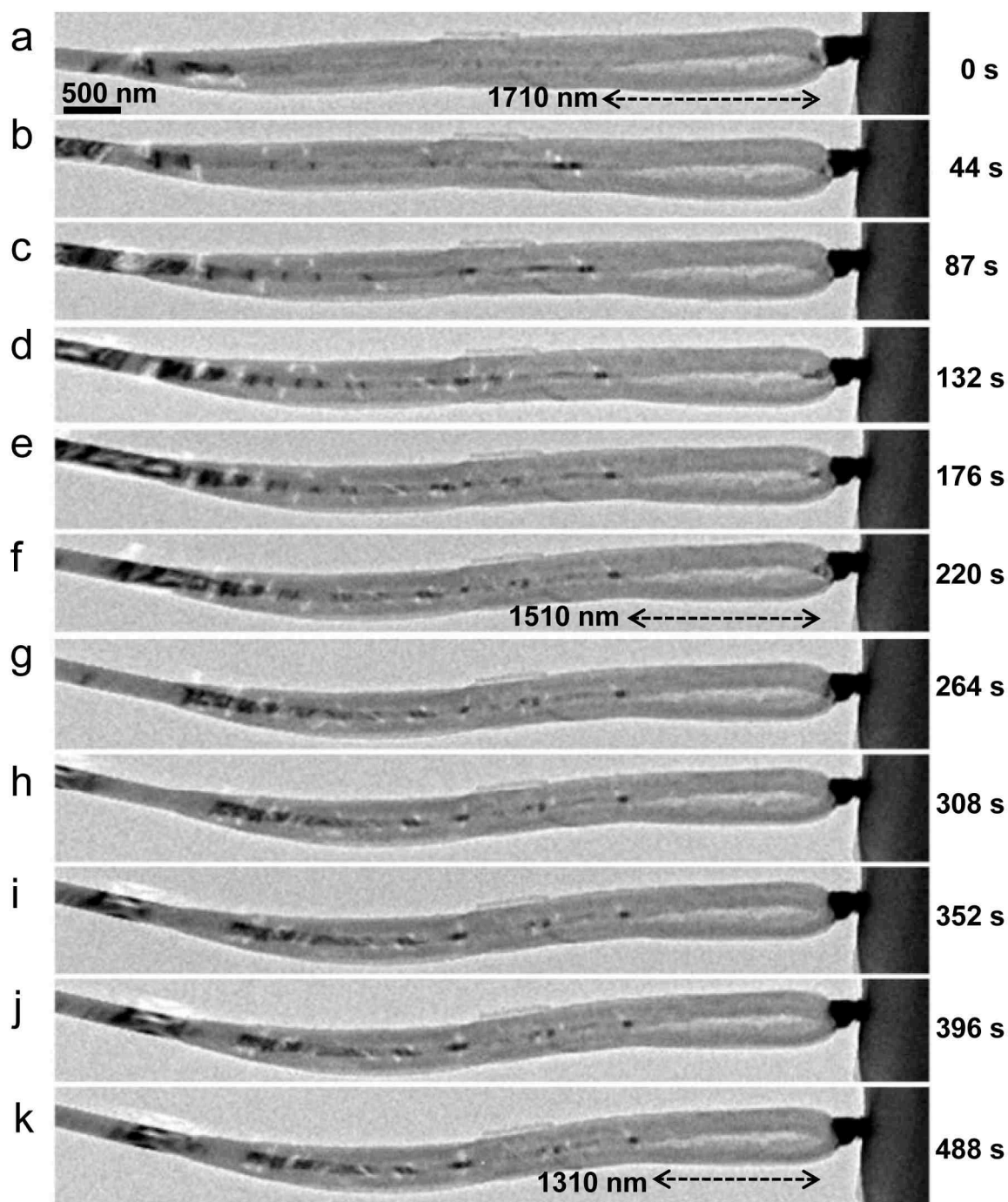


Figure S4. Crack and nanowire morphology evolution during delithiation of the same wire shown in Fig. 2. The nanowire was bent due to shrinkage after Li was extracted from the shell. The length of the crack also decreased from 1710 nm to 1310 nm in 8 minutes.

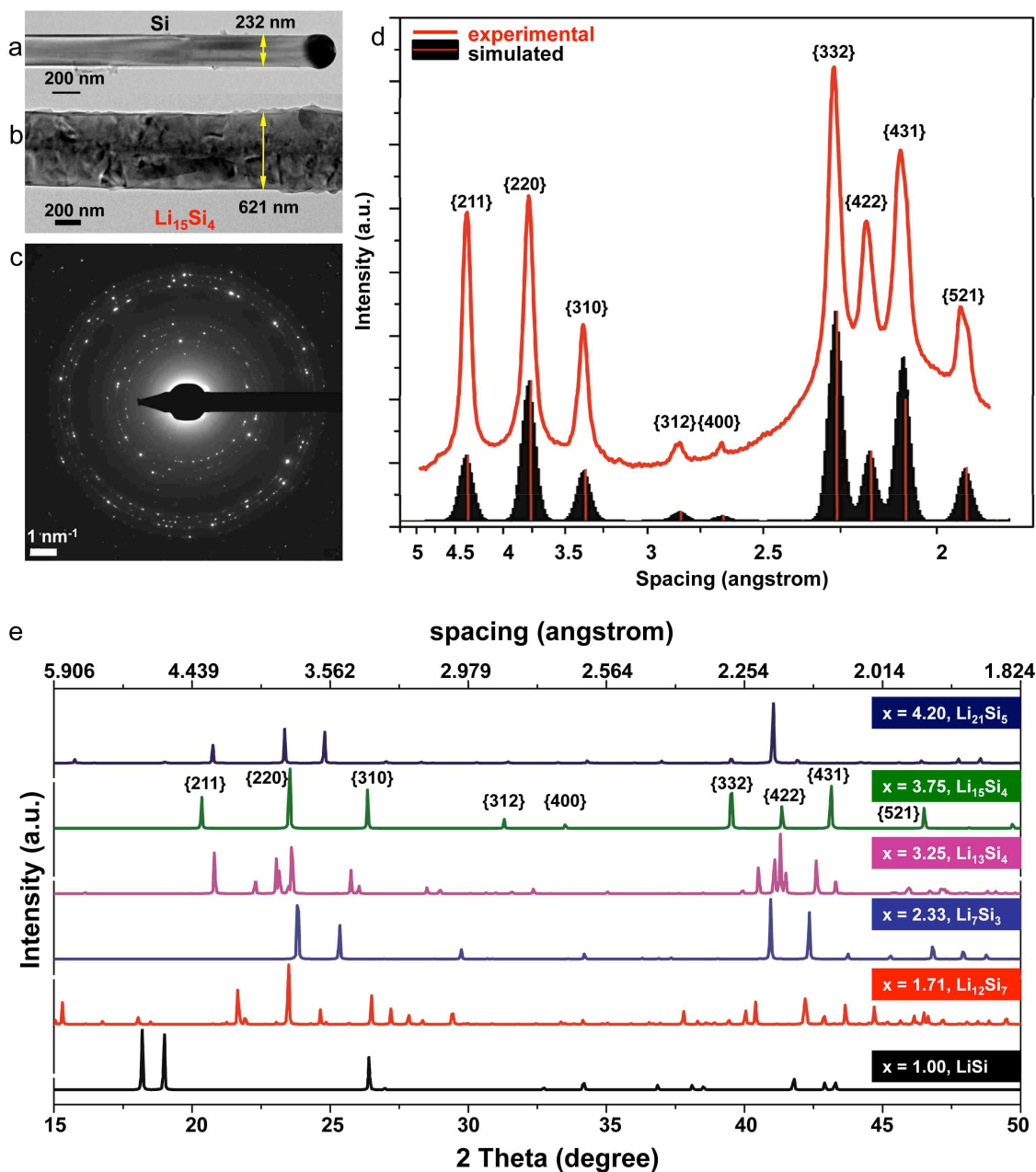


Figure S5. Identification of the $\text{Li}_{15}\text{Si}_4$ phase as the fully lithiated phase. (a) Pristine Si nanowire. (b-c) Lithiated Si nanowire (b) and its EDP (c). (d) Comparison of the intensity profiles of the experimental EDP and simulated EDP of the $\text{Li}_{15}\text{Si}_4$ phase, with excellent agreement for the nine observed strong diffraction rings. (e) Simulated X-ray diffraction patterns of the known Li-Si alloy phases with distinctive features. Note that for clarity, the pattern from the $\text{Li}_{22}\text{Si}_5$ phase is not presented here, because it resembles the pattern of the $\text{Li}_{21}\text{Si}_5$ phase due to almost identical atomic arrangement. The indices are only labeled for the observed $\text{Li}_{15}\text{Si}_4$ phase.

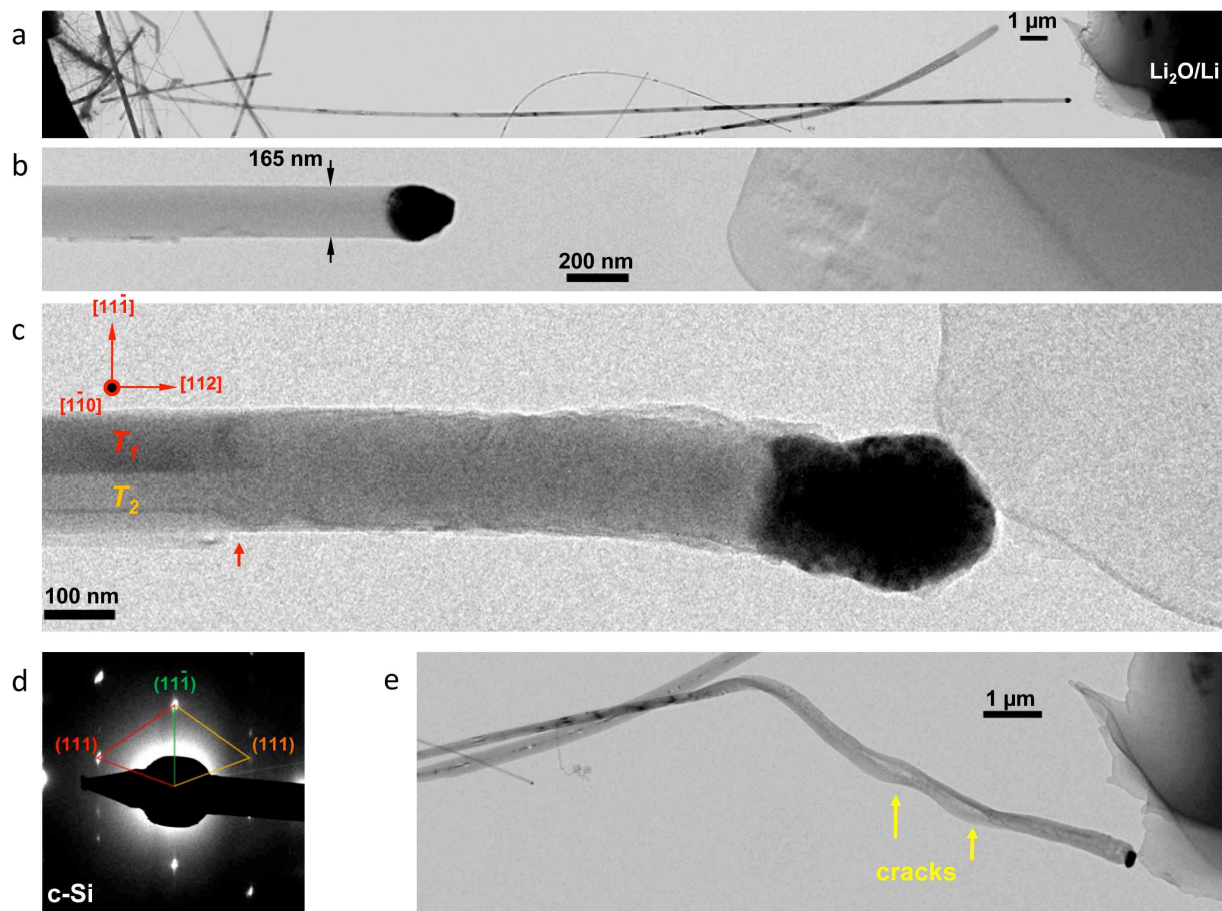


Figure S6. Lithiation of another [112]-oriented Si nanowire showing little expansion when viewed along the $[1\bar{1}0]$ direction. (a-b) Pristine Si nanowire. (c) Morphology of the nanowire in an intermediate state of lithiation. The red arrow marks the reaction front. A twin boundary was visible on the left side in the unreacted portion of the Si nanowire. Different image contrast on the left and right side of the reaction front is noticeable. The Au cap was elongated due to lithiation. Little expansion was seen along the $[11\bar{1}]$ direction. (d) Electron diffraction pattern from the pristine Si nanowire revealing the twin structure, consistent with the orientation relation shown in (c). (e) Tilted image showing cracks and a belt morphology of the lithiated nanowire.

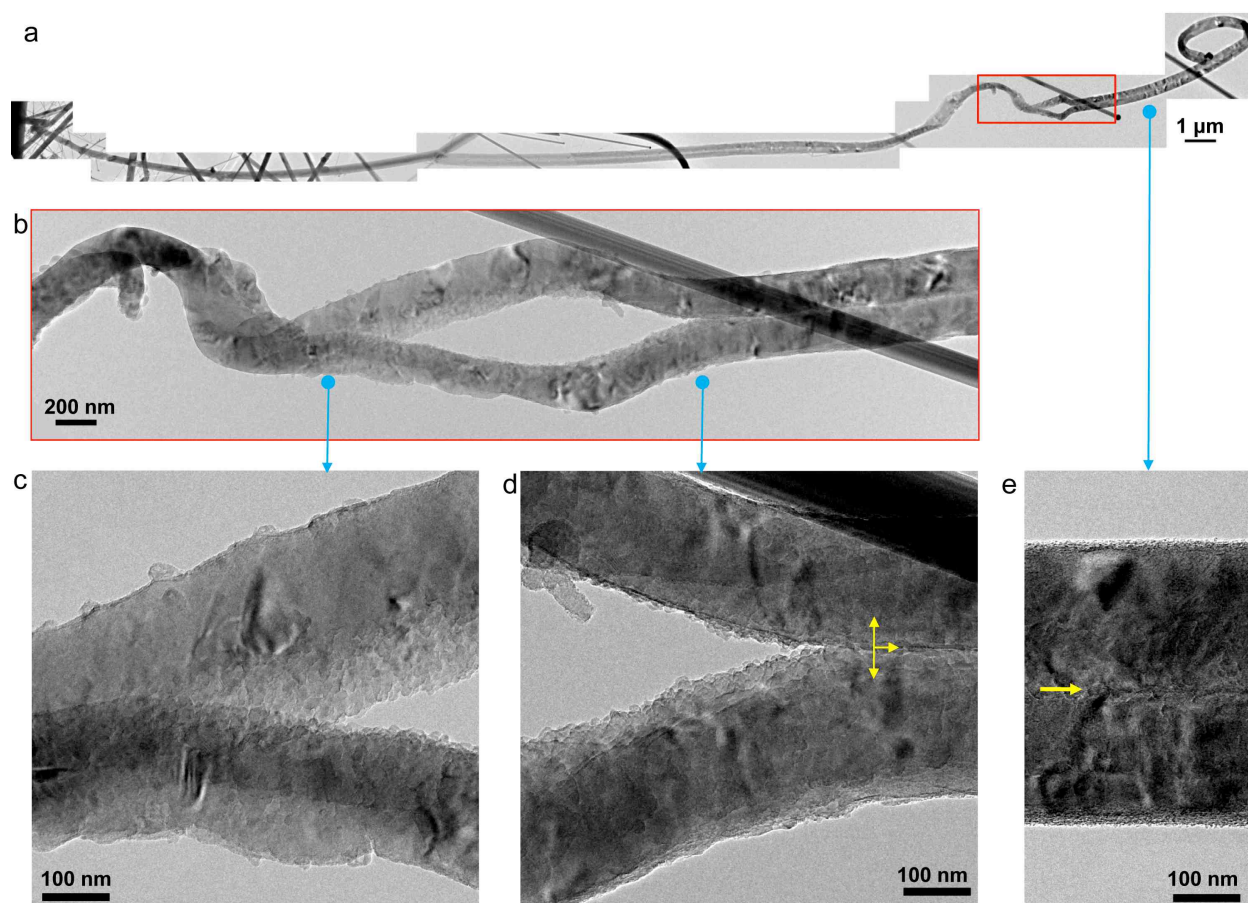


Figure S7. Self-splitting of a nanowire from the central crack during lithiation. (a) Low magnification TEM image of the lithiated nanowire. For this lithiation treatment, the potential for lithiation was -4 V with respect to Li metal. (b) Enlarged image showing the splitting/fracture of the nanowire. (c-d) Close-up views of the crease where the nanowire was split into two sub-wires. The yellow arrows mark the splitting directions and the central crack. (e) Non-split part with the central crack (marked by the yellow arrow).

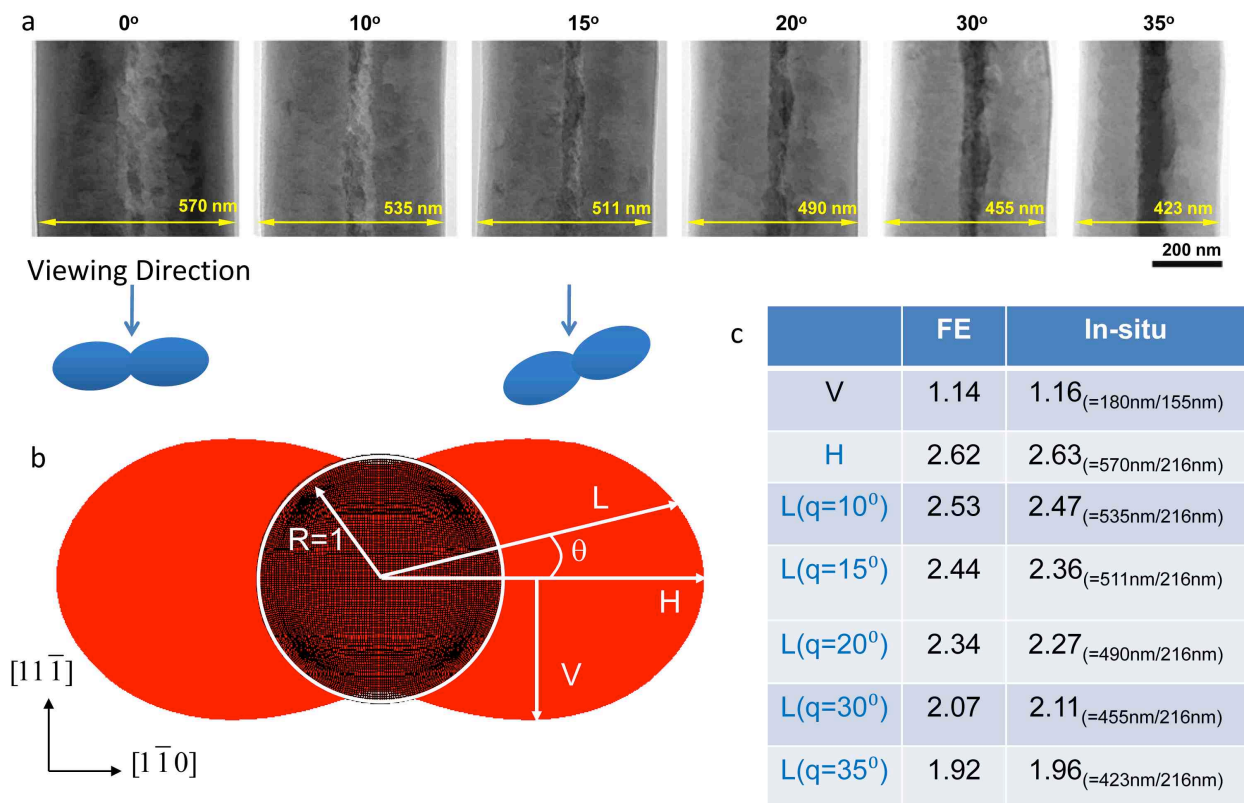


Figure S8. Comparison of the cross-sectional geometry between experiment and simulation.

(a) TEM images of the wire viewed from the side at different tilt angles. Because of the dumbbell-shaped cross section, the width of the wire, measured from the 2D projected view, changes with the viewing angle. Notice that the width of the Si core (in dark grey) increases with the viewing angle θ ; this indicates that the cross section of the Si core has an elliptical shape. (b) The simulated cross section of a fully lithiated nanowire, showing the anisotropic volume expansion and final dumbbell cross section after full lithiation. The initial cross section of a pristine Si nanowire is represented by the circle enclosing the finite element (FE) mesh and it is superposed to highlight the morphological changes. (c) Quantitative comparison of the cross-sectional geometry between finite element method simulation and in-situ TEM measurement, the table lists the ratios of the characteristic lengths L , V and H of the lithiated wire, defined in (b), to the initial diameter (216nm) of the pristine wire. The overall error is less than 4%.

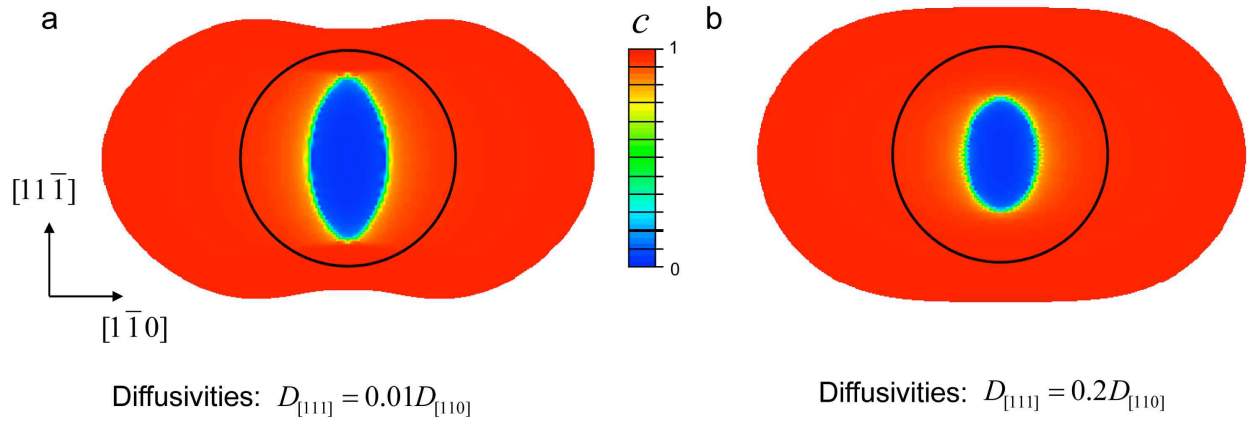


Figure S9. Sensitivity study of the Li diffusivity anisotropy on the necking instability. The Li distribution is compared when the diffusivity in the $[11\bar{1}]$ direction is (a) 0.01 times versus (b) 0.2 times that in the $[1\bar{1}0]$ direction. In case (a), the Si core is significantly elongated in the $[11\bar{1}]$ direction. In case (b), the elongation is significantly reduced, and the necking instability is consequently suppressed. Black circles indicate the initial cross section of the pristine Si nanowire.

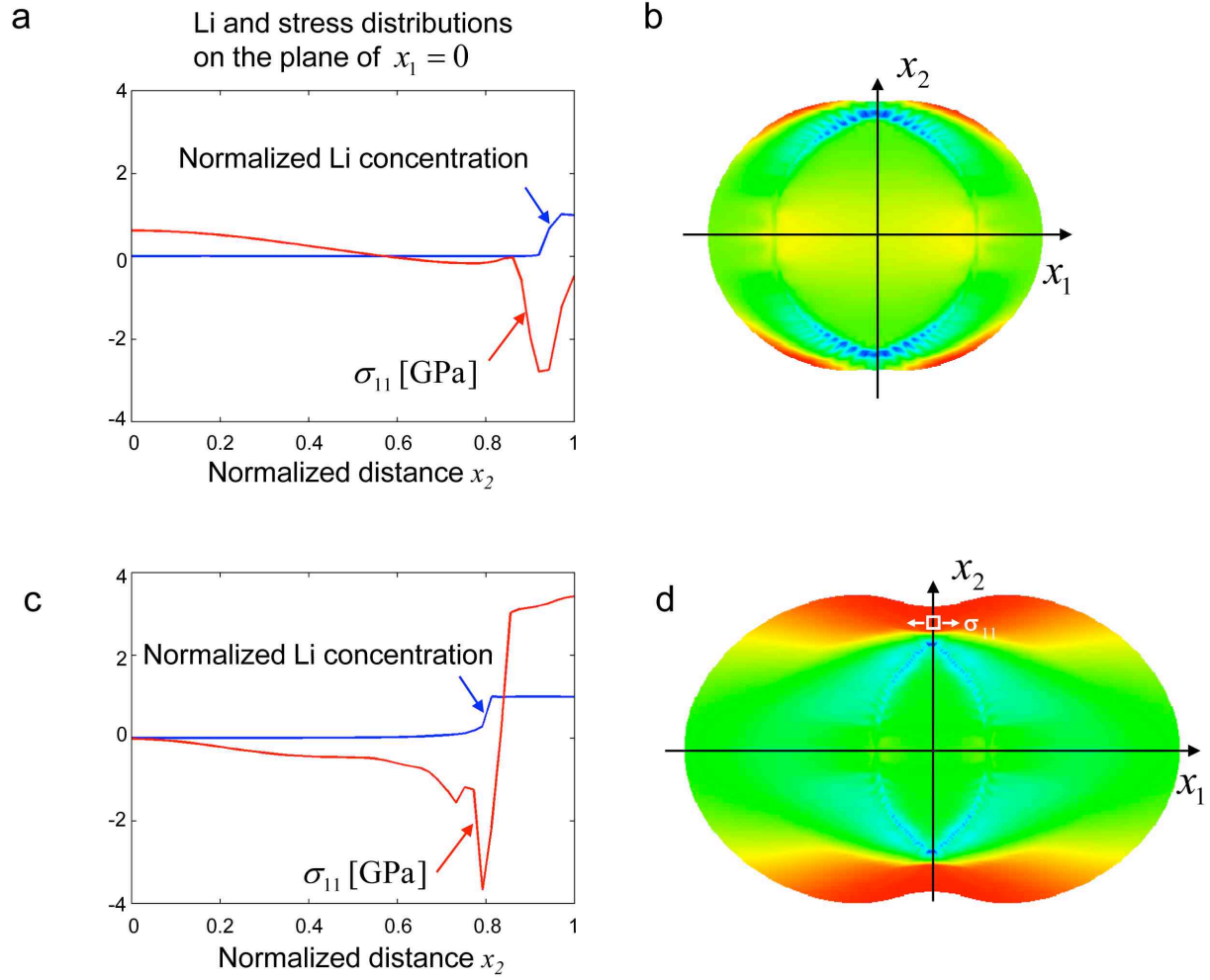


Figure S10. Distributions of Li and normal stress σ_{11} on the $x_2 - x_3$ plane at different deformation stages. (a-b) In the pre-necking stage, the lithiation reaction front (*i.e.* the region where the normalized Li concentration jumps from zero to one) is close to the surface, and σ_{11} is *compressive near the surface and tensile in the center*. (c-d) In the post-necking stage, the reaction front moves towards the center; σ_{11} becomes *tensile near the surface and compressive throughout the center*.

III. Supporting Movies

[nl201684d_si_002.mpg](#), [nl201684d_si_003.mpg](#), [nl201684d_si_004.mpg](#)

***In-situ* TEM movies showing the dumbbell cross-section of the lithiated Si nanowires.** The videos were recorded at 3 frames/s and played at 10× ([nl201684d_si_002.mpg](#)) and 50× ([nl201684d_si_003.mpg](#), [nl201684d_si_004.mpg](#)) speed.

[nl201684d_si_005.mpg](#), [nl201684d_si_006.mpg](#), [nl201684d_si_007.mpg](#)

***In-situ* TEM movies showing the lithiation of Si nanowires.** Note the volume expansion after the reaction front passed by and crack nucleation and propagation along the tail of the *c*-Si core. The videos were recorded at 3 frames/s and played at 50× ([nl201684d_si_005.mpg](#), [nl201684d_si_006.mpg](#)) and 100× ([nl201684d_si_007.mpg](#)) speed.

[nl201684d_si_008.mpg](#)

3D simulation of the Li distribution and anisotropic swelling in a progressively-lithiated Si nanowire.

[nl201684d_si_009.mpg](#)

Same as Movie [nl201684d_si_008.mpg](#), except that the wire is cut to expose the Li distribution in the (111) cross section, showing the development of the tapered core-shell structure along the longitudinal direction of the nanowire.

[nl201684d_si_010.mpg](#)

2D simulation of the cross-sectional Li distribution, showing the development of the core-shell structure and dumbbell-shaped cross section due to the onset and growth of a necking instability.

[nl201684d_si_011.mpg](#)

Same as Movie [nl201684d_si_010.mpg](#), except that the distribution of the von Mises equivalent stress is plotted, showing the development of extensive plastic flow due to progressive lithiation (red indicates attainment of the plastic yield stress).

IV. Identification of $\text{Li}_{15}\text{Si}_4$ as the final lithiated phase

The calculated structure factors (F) and extinction distances (ξ) of the observed diffracted beams are listed in **Table S1**. All the hkl diffraction rings are in excellent agreement with the experimental EDPs from the lithiated Si nanowires (**Fig. S5**). Kinetic approximation was valid because the smallest ξ for $\{332\}$ plane family is about 334.3 nm (highlighted in red in **Table S1**), larger than the thickness of the $\text{Li}_{15}\text{Si}_4$ nanowire, ~ 250 nm (**Figs. S4a-S4b**, note that the e-beam incident and view directions were close to $\langle 111 \rangle$ because **Fig. S4b** showed large diameter expansion).

The XRD patterns in **Fig. S4e** were generated with Materials Studio® using atomic coordinates of the six known Li_xSi phases listed in Dahn *et al.*'s publication.³ The widely assumed phase for fully lithiated Si, *i.e.* $\text{Li}_{22}\text{Si}_5$, is very similar to the $\text{Li}_{21}\text{Si}_5$ in atomic arrangement.⁴ As a result, the XRD patterns are very similar thus not piled in **Fig. S4e**.

Table S1. Structure factors and extinction distances of the observed 9 diffraction rings.

hkl	Spacing/ $\text{\AA}$	Structure Factor (F)	Amplitude of F , $ F ^2$	Multiplication of hkl	Extinction Distance, ξ/nm
211	4.350	-0.483-0.784 <i>i</i>	0.921	24	829.7
220	3.767	2.044+0.080 <i>i</i>	2.046	12	373.5
310	3.369	0.038-0.912 <i>i</i>	0.913	24	837.2
312	2.848	0.282-0.085 <i>i</i>	0.295	48	2593.5
400	2.664	0.606+0.029 <i>i</i>	0.607	6	1258.3
332	2.272	-1.463-1.756<i>i</i>	2.285	24	334.3
422	2.175	1.079+0.795 <i>i</i>	1.340	24	570.2
431	2.090	-0.608-0.480 <i>i</i>	0.774	48	986.7
521	1.945	-0.251-0.832 <i>i</i>	0.869	48	879.4

V. Description of modeling

We have performed both three-dimensional (3D) and two-dimensional (2D) simulations to explore the effects of lithiation on the morphology changes and fracture in Si nanowires. To simulate the development of the core-shell structure, we implemented a Li diffusion model with Li concentration-dependent diffusivities, facilitating the formation of a sharp interface that separates the Li-poor core (*i.e.*, the ratio of Li to Si $x \approx 0$) and Li-rich ($x \approx 3.75$) shell. Note that our simulations mainly serve to generate a sequence of core-shell structures for the mechanics analyses of deformation and fracture, rather than provide a precise description of the mechanisms of the dynamic lithiation process, which would be difficult due to lack of experimental measurements for model calibration.⁵⁻⁷ To this end, we employed a simple non-linear function on diffusivity

$$D = D_0 [1/(1-c) - 2\Omega c] \quad (1)$$

where D_0 is the diffusivity constant, c is the normalized Li concentration defined as the ratio between actual Li concentration divided by the Li concentration at the fully lithiated state ($x = 3.75$). On the basis of the free energy function of a regular solution model, $f = \Omega c(1-c) + [c \ln c + (1-c) \ln(1-c)]$, diffusivity D in Eq. (1) was derived according to a definition that can sharply increase diffusivities at high Li concentrations, $D = D_0 c d^2 f / dc^2$. In diffusion simulations, the normalized lithium concentrations behind the reaction front can quickly attain the high values (slightly below 1), while those ahead of the front remain nearly zero. This produces a sharp reaction front that is consistent with experimental observation, thereby providing a basis of further stress analysis. It should be noted that a small gradient of lithium concentration still exists behind the reaction front, so that Li can diffuse through the lithiated shell continuously to move forward the reaction front to the center of the wire. The non-linear relation of Eq. (1) is unbounded at $c = 1$; for numerical stability, the maximum of D is capped at $10^4 D_0$. In Eq. (1), the parameter Ω can be tuned to control the concentration profile near the reaction front, and we took $\Omega = 3$ for achieving the numerical stability.

In order to account for the influence of crystalline Si core on the migration velocity of the core-shell interface, the apparent Li diffusivity was assumed to be crystal orientation dependent. Such anisotropy plays a critical role in the formation of the dumbbell-shaped cross section, as numerically shown next. Among the non-zero direct components of diffusivity, D_{ij} , our numerical experiments indicated that D_{11} in the $[1\bar{1}0]$ direction should be about 100 times D_{22} in the $[11\bar{1}]$ direction. It should be emphasized that the assignment of anisotropic diffusivities serves as an effective numerical approach to capture the orientation-dependent migration velocity of the lithiation reaction front. Specifically, the anisotropic diffusivity in a “single phase” we used in our “diffusion” simulation was adopted to describe the anisotropic movements of what are actually two-phase interfaces. In reality, the two phases are $\alpha = \text{Si-Li alloy}$ in diamond cubic crystal structure (with Li fraction no more than 20% from TEM observations), and $\beta = \text{Li}_{15}\text{Si}_4 \text{ compound} + \text{Li}_2\text{O} + \text{amorphous Li}_x\text{Si}$ (for simplicity we lump them all together and pretend they are one “product” phase), characterized by much higher Li concentrations. Microscopy evidences indicate the α - β interface is both structurally and chemically very sharp, and moves by the atomic-ledge mechanism (to be published). Thus, the transport and chemical

reaction aspect of this problem can be naturally divided into three parts: (1) Li bulk diffusion in α , (2) Li bulk diffusion in β , and (3) motion of the $\alpha\beta$ interface where dramatic structural and chemical changes occur in the swept area. The physical origin of the anisotropy lies mostly in (3), even though (1) and (2) may have anisotropies also. The diffusivity model taken in our “single phase” simulation is just to mimic the anisotropic motion in (3) – as can be seen in Fig. 5, we indeed achieved rather sharp and moving interfaces that match with our experiments – the main purpose of our “diffusion” model. However, we would like to emphasize that the main point of our simulation is to compute the **stress distribution** due to lithiation transformation strain, that furthermore drives **plastic flow** and **fracture**, i.e. the mechanics part of the problem, not the transport and chemistry part. The rigorous framework for the transport and chemistry part is delayed to a future paper.

To mimic the fast Li diffusion at the surface,⁸ the Li surface diffusivity was taken as 300 times the bulk diffusivity of D_{11} , giving the tapered core-shell structure in 3D finite element simulations, consistent with experimental observation, as discussed later.

We adopted an elastic and perfectly plastic model to describe the lithiation-induced deformation. The increment of the total strain, $d\epsilon_{ij}$, is taken to be the sum of three contributions,⁹

$$d\epsilon_{ij} = d\epsilon_{ij}^c + d\epsilon_{ij}^e + d\epsilon_{ij}^p \quad (2)$$

In Eq. (2), the increment of the lithiation-induced chemical strain, $d\epsilon_{ij}^c$, is proportional to the increment of the normalized Li concentration, $d\epsilon_{ij}^c = \beta_{ij}dc$. Here β_{ij} represents the expansion coefficient with non-zero direct components, and they are controlled by the atomic processes near the lithiation reaction front and are expected to be anisotropic for lithiation of c -Si. Since a mechanistically-based assignment of β_{ij} is beyond the scope of this work, we performed numerical experiments to determine β_{ij} by adjusting β_{ij} to give a close fit of the measured geometry of fully lithiated Si nanowires. By this procedure, we obtained $\beta_{11} = 150\%$ in $\langle 110 \rangle$, $\beta_{22} = 40\%$ in $\langle 111 \rangle$, and $\beta_{33} = 0$ in $\langle 112 \rangle$ directions.

In Eq. (2), the increment of the elastic strain, $d\epsilon_{ij}^e$, obeys Hooke’s law, $d\epsilon_{ij}^e = [(1+\nu)d\sigma_{ij} - \nu d\sigma_{kk}\delta_{ij}]/E$, where E is Young’s modulus and ν is Poisson’s ratio, $\delta_{ij} = 1$ when $i = j$, $\delta_{ij} = 0$ otherwise, and the repeated index means summation. Our elasticity model accounts for the Li concentration-dependent E and ν ; for pristine Si, $E = 80$ GPa, $\nu = 0.28$, and for fully lithiated Si, $E = 30$ GPa, $\nu = 0.24$; both E and ν linearly vary between the two limits as suggested by recent first-principles calculations.¹⁰

In Eq. (2), the increment of the plastic strain, $d\epsilon_{ij}^p$, was assumed to obey the classic J_2 -flow rule; namely plastic yielding occurs when the von Mises equivalent stress, $\sigma_{eq} = \sqrt{3s_{ij}s_{ij}/2}$, equals the yield stress σ_Y . Here $s_{ij} = \sigma_{ij} - \sigma_{kk}\delta_{ij}/3$ is the deviatoric stress; $d\epsilon_{ij}^p$ is proportional to s_{ij} and can be determined by solving the boundary value problem.

We numerically implemented the above diffusion and elastic-plastic model by using the finite element package ABAQUS 6.10.¹¹ The 3D simulation can well capture the overall wire deformation and development of the tapering core-shell structure, while the 2D cross-sectional simulation (with fine meshes) enables the quantification of geometry of a fully lithiated nanowire, which is consistent with experimental measurements. The Li and stress-strain fields were incrementally updated using an implicit coupled temperature-displacement procedure in ABAQUS/Standard. Specifically, the lithiation-induced strain was simulated by a thermal strain approach. That is, the normalized concentration was surrogated by temperature and the expansion coefficient β_{ij} was equivalently treated as the thermal expansion coefficient. The user material subroutine for heat transfer (UMATHT) was programmed to interface with ABAQUS for updating diffusivities based on the current Li concentration (*i.e.*, temperature). The Li distribution was then updated and so was the increment of the elastic-plastic deformation. In order to numerically stabilize the large distortions at both the reaction front and growing neck, a small rate sensitivity was used by assigning the exponent of 500 for the over-stress power law model of rate-dependent viscoplasticity.¹¹

In 2D cross-sectional simulations, the plane-strain condition was assumed on the basis of the negligible axial deformation observed in experiments. We used the 4-node plane strain element with thermally coupled quadrilateral, bilinear displacement and temperature (CPE4T). The in-plane boundary conditions were taken as a constant Li concentration ($c = 1$) and zero tractions at the outer boundary. Utilizing the symmetry condition, one quarter of the wire was modeled to reduce the computation cost. For 3D simulations, the Li surface flux was prescribed at one end of the wire and a skin layer was additionally attached to the wire surface. As mentioned earlier, to mimic the fast Li diffusion at the surface, the Li diffusivity in this skin layer was taken as 300 times that in the bulk. The simulation of development of the core-shell structure (**Movie S8**) agrees with *in situ* TEM observations. In addition, the 3D simulation of distributions of the Li concentration and stresses are consistent with the results from 2D cross-sectional simulations. Such a close correlation can be understood by noting that our non-linear diffusivity model (**Eq. 1**) essentially forces the normalized Li concentrations to be close to either zero (pristine Si) or one (fully lithiated Si). As a consequence, the in-plane diffusion in 3D simulations occurs mostly under the condition of a fully-lithiated surface, which is the boundary condition in 2D simulations. Both our 2D and 3D simulations well captured the main features of the morphology changes in experiments.

We have performed numerical experiments to evaluate the sensitivity of our simulations to assigned Li diffusivities. Of particular note is the influence of the anisotropy of apparent Li diffusivities, which characterizes the orientation-dependent migration velocity of the lithiation reaction front, as discussed earlier. **Figure S9** shows that an increase of such anisotropy can promote the elongation of the shrinking Si core in the $[11\bar{1}]$ direction, which is fundamentally needed to facilitate the necking instability. Specifically, in reference to the coordinate system in **Fig. 5a** of the text, let us focus on the lithiated shell near the apex of the cross section. As lithiation proceeds, if the thickness of such a shell, measured in the x_2 direction of $[11\bar{1}]$, can increase significantly with x_1 , it will generate a large variation of the outward surface displacement, which increases with x_1 . As a result, the top surface will gradually become flat and even slightly indented. The ensuing stress concentration at the indent facilitates the onset of

necking instability. Further post-necking deformation can lead to the formation of the observed dumbbell-shaped cross section.

The physical origin of the necking instability can be revealed by analyzing the elastic-plastic deformations of the surface layer near the apex of the cross section. Upon lithiation, a compressive hoop stress first develops in this layer, due to the circumferential constraint on volume expansion (Figs. S10a, S10b). This is followed by the compressive plastic yielding with a further increase of the local Li concentration. The layer becomes fully lithiated after the passing of the reaction front. As lithiation proceeds, the volume expansion at the lithiation front displaces the materials behind it in the outward radial direction and simultaneously stretches them in circumferential directions. As a result, the surface layer sequentially undergoes *compressive elastic unloading*, *tensile elastic loading* and *tensile plastic yielding* (Figs. 5e-5f and Figs. S10c, S10d). This is reminiscent of the deformation sequence at the crack tip under cyclic mechanical load.¹² It should be emphasized that the tensile plastic flow is only a necessary condition of a necking instability. Our simulations (Movies S7-S10) clearly show the non-uniform surface displacement near the apex of the cross section during the pre-necking deformation, mainly caused by the lithiation strain in the $[11\bar{1}]$ direction and promoted by the $[11\bar{1}]$ -elongated Si core (Figs. S9a, 5d). Consequently, the surface near the apex gradually becomes flat and slightly indented. Driven by the progressive lithiation of the Si core, further expansion of the lithiated shell causes the onset of a necking instability around the surface indent, followed by neck growth and formation of the dumbbell-shaped cross section, consistent with experimental observation.

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